KAIZEN MATHS

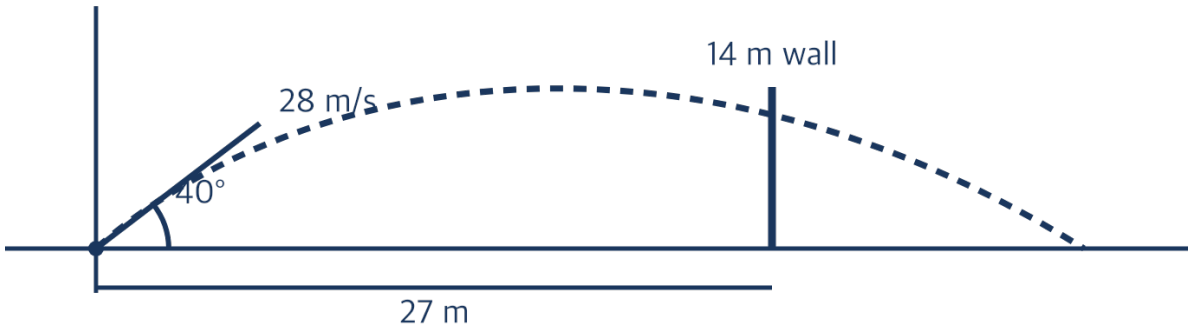
Lesson starter

Projectile Motion

Modelling Problems

Name _____

SA1



A projectile is launched at 28 m/s at 40°. A wall of height 14 m is 27 m from the launch point. Determine whether the projectile clears the wall.

MEC-5B63-L3 · SA

Show your working

KAIZEN MATHS

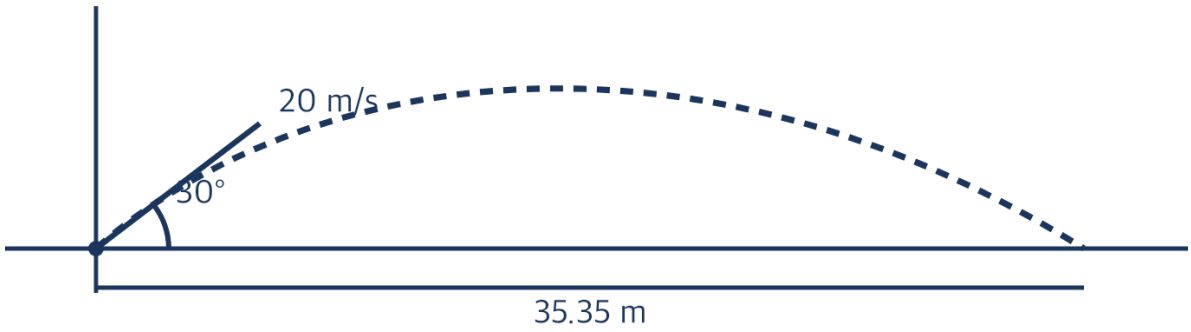
Lesson starter

Projectile Motion

Modelling Problems

Name _____

SB1



A projectile is launched from level ground at an angle of 30°. It lands 35.35 m away. Find its initial speed.

MEC-5B63-L3 · SB

Show your working

KAIZEN MATHS

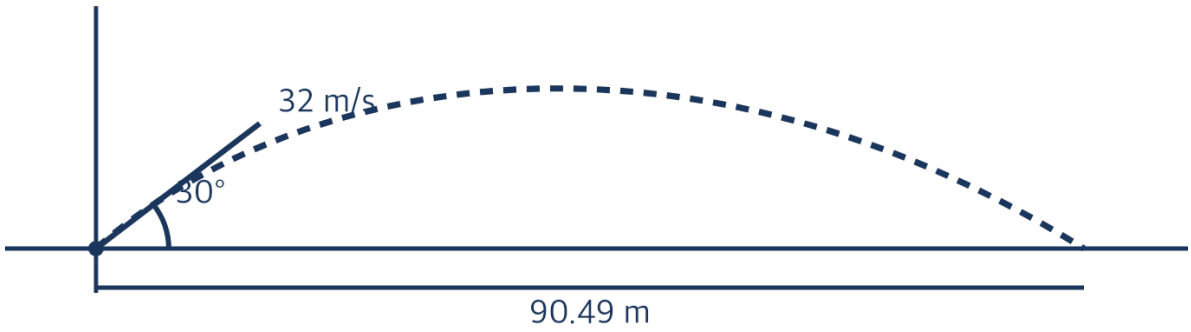
Lesson starter

Projectile Motion

Modelling Problems

Name _____

SC1



A projectile is launched from level ground with speed 32 m/s and lands 90.49 m away. Find the possible angles of projection.

MEC-5B63-L3 · SC

Show your working

KAIZEN MATHS

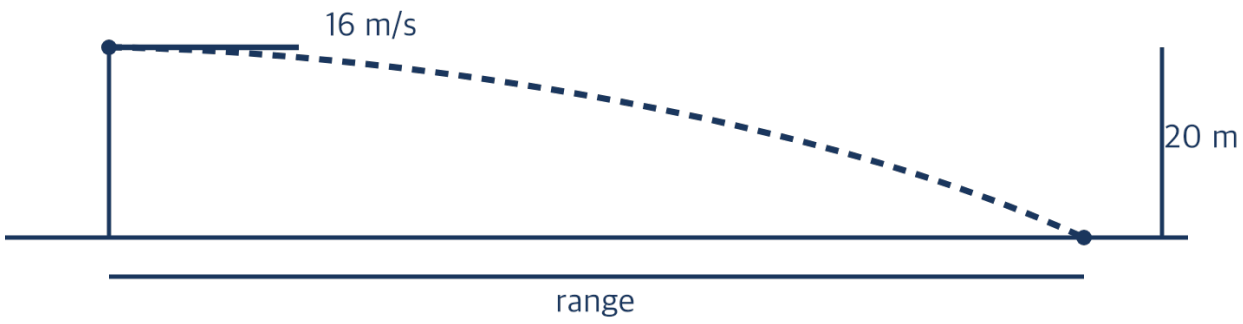
Lesson starter

Projectile Motion

Modelling Problems

Name _____

SD1



A ball is projected horizontally from the top of a cliff 20 m high with speed 16 m/s. Find the time taken to hit the ground and the horizontal distance travelled.

MEC-5B63-L3 · SD

Show your working

KAIZEN MATHS

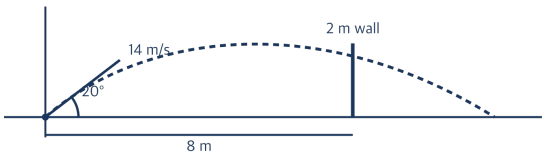
Exit ticket

Projectile Motion

Modelling Problems

Name _____

EA1



A projectile is launched at 14 m/s at 20°. A wall of height 2 m is 8 m from the launch point. Determine whether the projectile clears the wall.

Show your working

KAIZEN MATHS

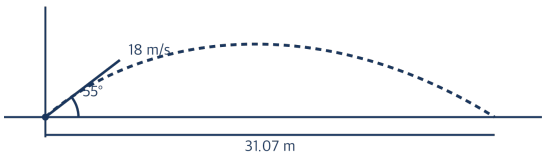
Exit ticket

Projectile Motion

Modelling Problems

Name _____

EB1



A projectile is launched from level ground at an angle of 55°. It lands 31.07 m away. Find its initial speed.

Show your working

KAIZEN MATHS

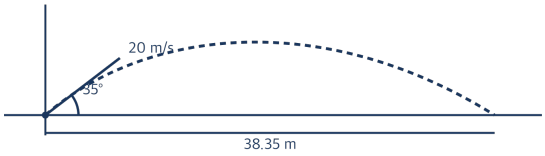
Exit ticket

Projectile Motion

Modelling Problems

Name _____

EC1



A projectile is launched from level ground with speed 20 m/s and lands 38.35 m away. Find the possible angles of projection.

Show your working

KAIZEN MATHS

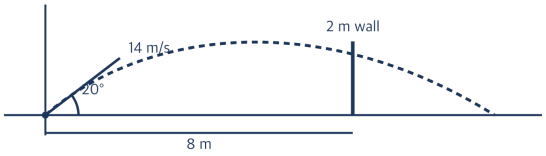
Exit ticket

Projectile Motion

Modelling Problems

Name _____

EA1



A projectile is launched at 14 m/s at 20°. A wall of height 2 m is 8 m from the launch point. Determine whether the projectile clears the wall.

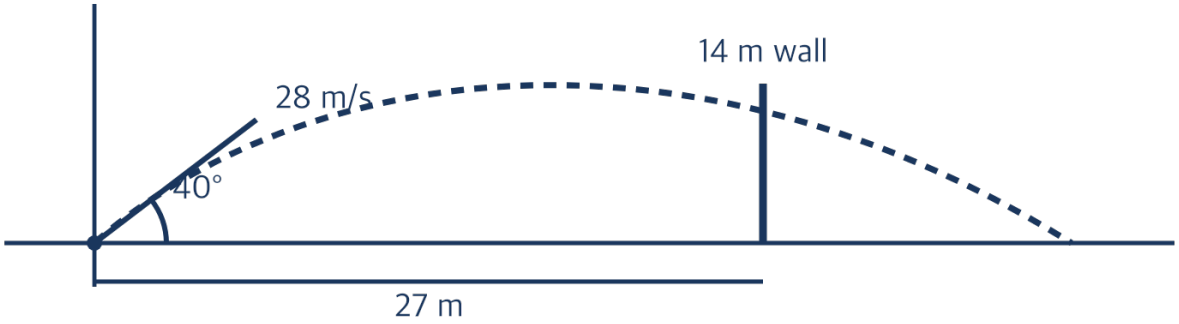
Show your working

Teacher answers & working

Projectile Motion · Modelling Problems

MEC-5B63-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



A projectile is launched at 28 m/s at 40°. A wall of height 14 m is 27 m from the launch point. Determine whether the projectile clears the wall.

Answer:

clears the wall

Working:

Resolve the velocity.

$u_x = 21.45$ m/s, $u_y = 18$ m/s

Use horizontal motion to find the time taken to reach the wall.

$t = 27 \div 21.45 = 1.26$ s

Find the projectile height at that time.

$y = 18 \times 1.26 - 4.9 \times 1.26^2 = 14.89$ m

Compare the projectile height with the wall height.

$14.89 \text{ m} > 14 \text{ m}$

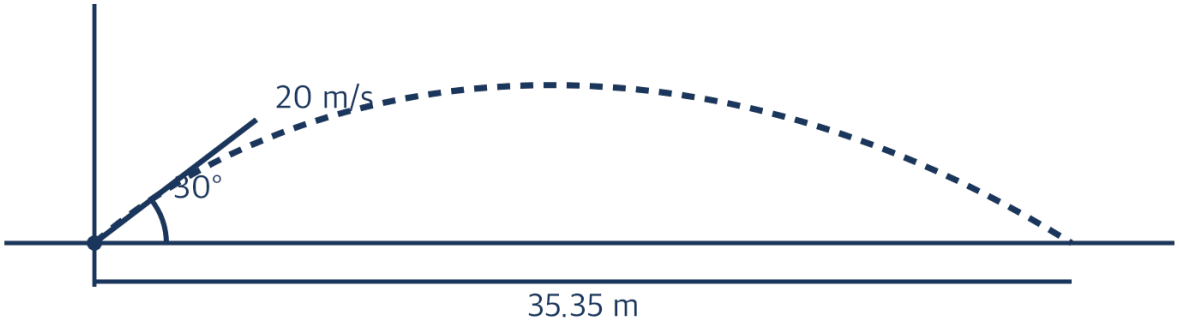
The projectile clears the wall.

Teacher answers & working

Projectile Motion · Modelling Problems

MEC-5B63-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1



A projectile is launched from level ground at an angle of 30°. It lands 35.35 m away. Find its initial speed.

Answer:

$u = 20\ \text{m/s}$

Working:

Use the range formula for launch and landing at the same height.

$R = u^2 \sin(2\theta) \div g$

Rearrange for u^2.

$u^2 = Rg \div \sin(2\theta)$

Substitute R = 35.35, θ = 30° and g = 9.8.

$u^2 = 35.35 \times 9.8 \div \sin(60°) = 400$

$u = \sqrt{400} = 20$

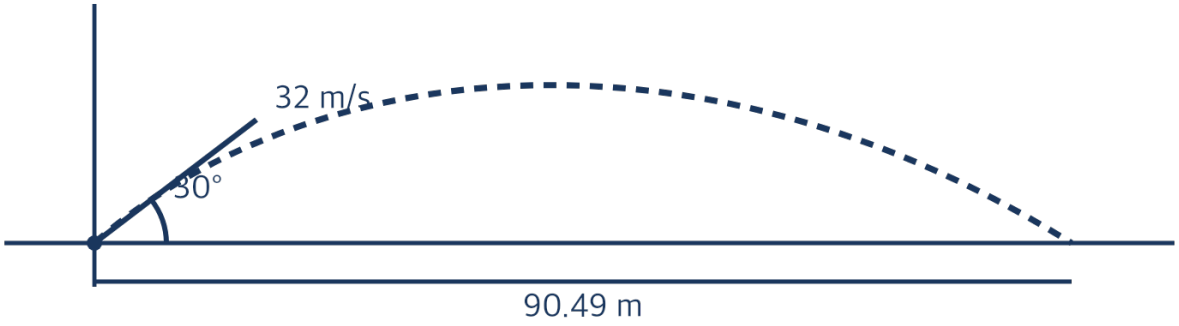
The initial speed is 20 m/s.

Teacher answers & working

Projectile Motion · Modelling Problems

MEC-5B63-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1



A projectile is launched from level ground with speed 32 m/s and lands 90.49 m away. Find the possible angles of projection.

Answer:

$\theta = 30°$ or $60°$

Working:

Use the range formula for level ground.

$R = u^2 \sin(2\theta) \div g$

Rearrange to find $\sin(2\theta)$.

$\sin(2\theta) = Rg \div u^2$

Substitute the values.

$\sin(2\theta) = 90.49 \times 9.8 \div 32^2 = 0.866$

Find the two possible values of 2θ between 0° and 180°.

$2\theta = 60°$ or $120°$

$\theta = 30°$ or $60°$

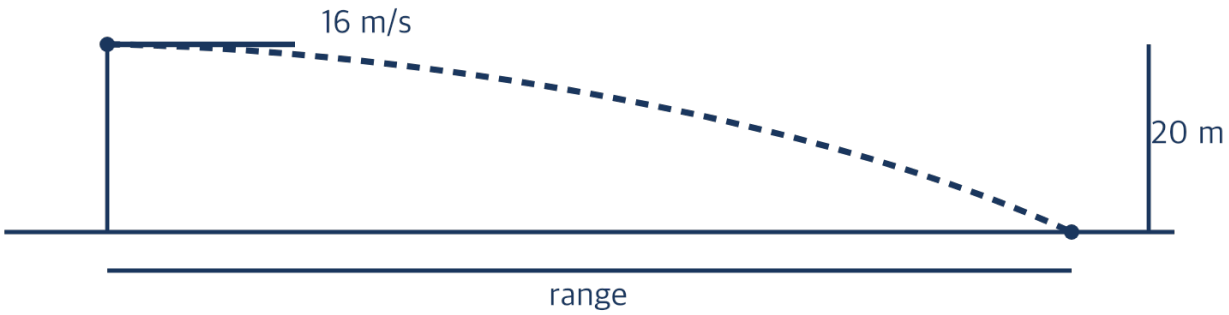
The possible projection angles are 30° and 60°.

Teacher answers & working

Projectile Motion · Modelling Problems

MEC-5B63-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1



A ball is projected horizontally from the top of a cliff 20 m high with speed 16 m/s. Find the time taken to hit the ground and the horizontal distance travelled.

Answer:

t = 2.02 s, distance = 32.32 m

Working:

Vertical motion starts with vertical velocity 0 and displacement -20 m if upward is positive.

$-20 = 0 \times t - 4.9t^2$

$4.9t^2 = 20$

$t^2 = 20 \div 4.9 = 4.08$

$t = 2.02$ s

Horizontal velocity is constant.

horizontal distance = 16 × 2.02 = 32.32 m

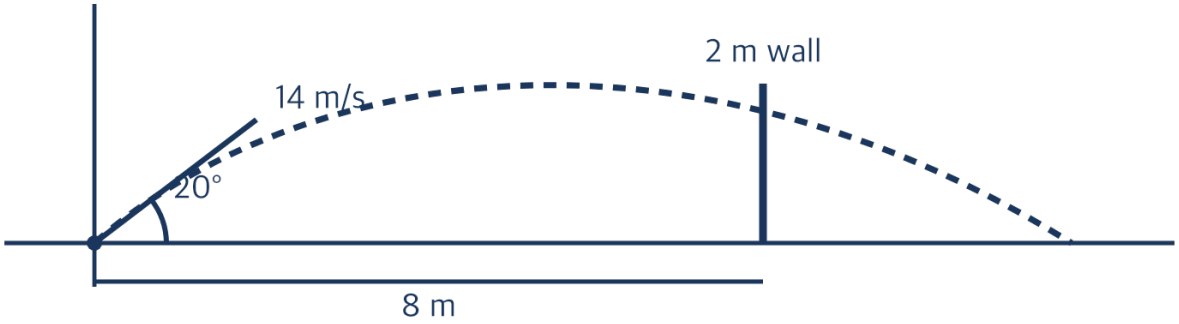
The time is 2.02 s and the horizontal distance is 32.32 m.

Teacher answers & working

Projectile Motion · Modelling Problems

MEC-5B63-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1



A projectile is launched at 14 m/s at 20°. A wall of height 2 m is 8 m from the launch point. Determine whether the projectile clears the wall.

Answer:

does not clear the wall

Working:

Resolve the velocity.

$u_x = 13.16$ m/s, $u_y = 4.79$ m/s

Use horizontal motion to find the time taken to reach the wall.

$t = 8 \div 13.16 = 0.61$ s

Find the projectile height at that time.

$y = 4.79 \times 0.61 - 4.9 \times 0.61^2 = 1.1$ m

Compare the projectile height with the wall height.

$1.1 \text{ m} \leq 2 \text{ m}$

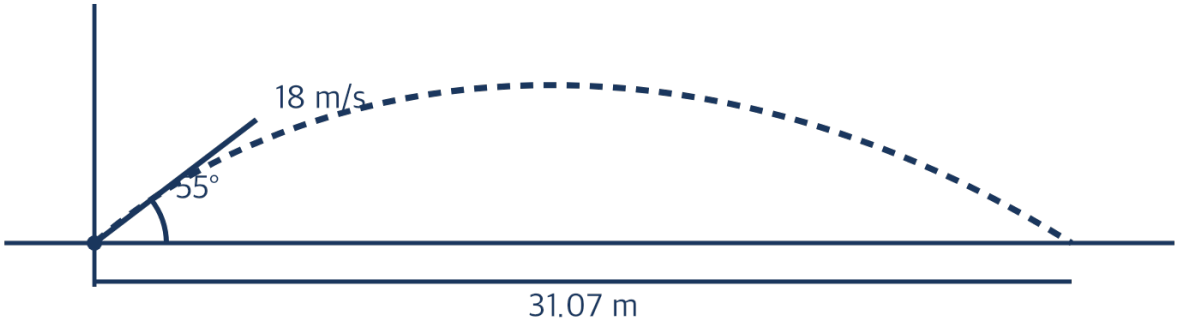
The projectile does not clear the wall.

Teacher answers & working

Projectile Motion · Modelling Problems

MEC-5B63-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1



A projectile is launched from level ground at an angle of 55°. It lands 31.07 m away. Find its initial speed.

Answer:

$u = 18$ m/s

Working:

Use the range formula for launch and landing at the same height.

$R = u^2 \sin(2\theta) \div g$

Rearrange for u^2.

$u^2 = Rg \div \sin(2\theta)$

Substitute R = 31.07, θ = 55° and g = 9.8.

$u^2 = 31.07 \times 9.8 \div \sin(110°) = 324$

$u = \sqrt{324} = 18$

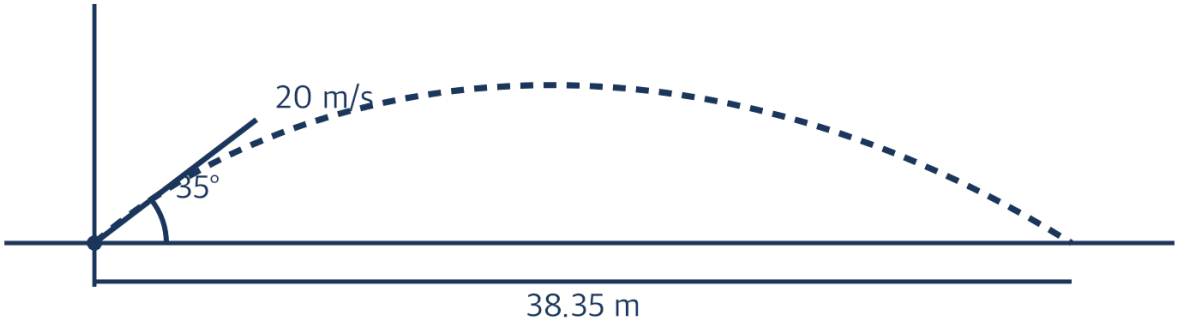
The initial speed is 18 m/s.

Teacher answers & working

Projectile Motion · Modelling Problems

MEC-5B63-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1



A projectile is launched from level ground with speed 20 m/s and lands 38.35 m away. Find the possible angles of projection.

Answer:

$\theta = 35°$ or $55°$

Working:

Use the range formula for level ground.

$R = u^2 \sin(2\theta) \div g$

Rearrange to find $\sin(2\theta)$.

$\sin(2\theta) = Rg \div u^2$

Substitute the values.

$\sin(2\theta) = 38.35 \times 9.8 \div 20^2 = 0.94$

Find the two possible values of 2θ between 0° and 180°.

$2\theta = 70°$ or $110°$

$\theta = 35°$ or $55°$

The possible projection angles are 35° and 55°.